

From Prediction to Action: Customer Lifetime Value Estimation and CRM Segmentation Using BG/NBD, Gamma-Gamma, and K-Means Clustering

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Course: DDM – Data-Driven Marketing

Abstract

This study develops a data-driven framework for revenue-based customer lifetime value (CLV) estimation and customer segmentation in a non-contractual retail setting. Using a large-scale retail transaction dataset, customer-level behavioral variables are constructed from transaction history, including recency, frequency, observation period, and monetary value. The BG/NBD model is applied to estimate expected future purchase periods and customers' probability of being alive, while the Gamma-Gamma model is used to estimate expected average transaction value. These outputs are combined to calculate six-month revenue-based CLV. K-means clustering is then applied using predicted CLV and probability of being alive to classify repeat customers into four strategic segments: Lost, At-risk, Potential, and VIP. The updated validation results show that BG/NBD provides useful holdout performance for purchase-frequency prediction and that p_alive meaningfully ranks future activity. The Gamma-Gamma diagnostic indicates limited individual-level accuracy in predicting average transaction value; therefore, CLV is interpreted as a ranking and segmentation indicator rather than an exact customer-level revenue forecast. Segment-level holdout validation shows clear separation in actual future activity and value, supporting the practical usefulness of the four CRM segments. Because the dataset does not include profit margin, cost-to-serve, or campaign response data, the proposed marketing strategies should be viewed as decision-support recommendations that require further validation through A/B testing, uplift modeling, and incremental CLV analysis before implementation.

Keywords: BG/NBD, customer segmentation, revenue-based CLV, Gamma-Gamma, K-means clustering.

1. Introduction

In increasingly competitive markets, rising acquisition costs have shifted managerial focus from transaction-oriented marketing toward relationship-based strategies that emphasize customer retention and value maximization (Kumar & Reinartz, 2012; Venkatesan & Kumar, 2004). Because modern consumers differ substantially in their needs and value potential, effective customer segmentation is critical. It allows firms to divide their customer base into homogeneous groups, prioritize marketing actions, and efficiently allocate resources (Wedel & Kamakura, 2000; Yao et al., 2014; Webster, 1992). Traditionally, the RFM (recency, frequency, monetary value) framework has been widely used for segmentation due to its simplicity and practical applicability (Fader et al., 2005; Bauer, 1988). However, RFM-based approaches are inherently backward-looking, relying on historical transactions, and may fail to fully capture customers' future value potential or their likelihood of remaining active. To address these shortcomings, this study adopts a forward-looking approach based on customer lifetime value (CLV). The BG/NBD model is applied to estimate expected future purchase frequency and the probability of being alive, while the Gamma-Gamma model estimates expected monetary value. By combining these outputs, customers are segmented into strategic groups reflecting both economic value and relationship status. This study contributes to data-driven marketing by translating probabilistic CLV predictions into actionable segmentation (identifying high-value, potential, at-risk, and

inactive customers), providing a practical foundation for targeted CRM strategies. The remainder of this paper outlines the literature (Section 2), methodology (Section 3), empirical results (Section 4), and conclusions (Section 5).

2. Literature review

2.1. Customer lifetime value and segmentation

Customer segmentation is a fundamental activity in marketing and has become increasingly important as firms seek to gain a deeper understanding of customers' characteristics, needs, and behavioral differences. The concept of customer segmentation was first introduced by Smith (1956) and has since been widely supported in the marketing literature (Alves Gomes & Meisen, 2023). In general, customer segmentation refers to the process of dividing a heterogeneous customer base into smaller and more homogeneous groups whose members share similar needs, characteristics, or behavioral patterns (McDonald & Dunbar, 2004).

Customer segmentation provides two major benefits for marketing management. First, it enables firms to identify strategically important customer groups, particularly those that are more profitable, loyal, or responsive to marketing actions (Dibb, 1998; Wind, 1978). Prior studies have used segmentation to evaluate customers' profit potential and support customer prioritization decisions (Storbacka, 1997; Mark, 2012). This allows managers to allocate marketing resources more efficiently by focusing on customer groups that are expected to generate greater value. Second, segmentation helps firms better understand customer behavior, preferences, and needs across different groups. Such insights allow firms to design more relevant products, services, communication messages, and relationship strategies for each customer segment (Dibb, 1998; Dibb & Simkin, 1997). Accordingly, several studies have applied customer segmentation to develop differentiated customer management and marketing strategies for distinct customer groups (Chan, 2008; Li et al., 2011; Wei et al., 2012, 2013).

In segmentation research, customers can be grouped based on different types of variables, which are commonly classified into general variables and product-specific variables (Wedel & Kamakura, 2000). General variables include demographic and lifestyle characteristics, such as age, gender, income, education, and lifestyle. Product-specific variables, in contrast, include purchase-related indicators such as purchase frequency, consumption behavior, spending level, and purchase intention. Although general variables are often easier to collect and apply, product-specific variables are more effective in capturing actual customer behavior and are therefore more useful for evaluating customers' economic contribution to the firm (Tsai & Chiu, 2004).

Traditional segmentation approaches rely heavily on product-specific behavioral variables such as recency, frequency, and monetary value (Apichottanakul et al., 2021; Heldt et al., 2021). These methods are popular because of their simplicity, interpretability, and practical applicability. However, RFM-based segmentation and its variants are primarily backward-looking, they classify customers based on historical transaction patterns rather than expected future contribution (Peker et al. 2017; Gan et al., 2024). As a result, traditional segmentation may not fully capture customers' future profitability or their likelihood of remaining active.

Customer lifetime value addresses this limitation by providing a forward-looking measure of customer profitability. CLV estimates the expected discounted future value generated by a customer, making it a useful metric for customer prioritization, retention management, and marketing resource allocation (Gupta et al., 2004; Venkatesan & Kumar, 2004). From a CRM perspective, CLV allows firms to manage customers as long-term assets rather than isolated transactions, thereby supporting more strategic customer relationship decisions (Kumar & Reinartz, 2018). Therefore, CLV-based segmentation provides a stronger foundation for customer classification because it considers not only past behavior, but also future value potential. This enables firms to identify high-value customers, allocate marketing resources more efficiently, and design differentiated retention strategies.

2.2. Probabilistic models in CLV

Accurate CLV estimation is challenging because customer behavior is uncertain and heterogeneous. Customers vary not only in how often they purchase, but also in how much they spend and whether they remain active over time. This challenge is especially important in non-contractual settings, where firms cannot directly observe customer churn (Mccarthy & Fader, 2018). A customer may stop purchasing temporarily or may have permanently left the firm, but this status is not explicitly

recorded.

Probabilistic models address this issue by modeling customer purchase behavior as a stochastic process. In particular, buy-till-you-die models assume that customers continue to make purchases while they remain active, but may become inactive at an unobserved point in time (Schmittlein et al., 1987; Kim et al., 2022). These models are useful because they estimate both future transaction frequency and the probability that a customer is still active.

The BG/NBD model is used for predicting repeat purchases in non-contractual contexts using customer-level information such as frequency, recency, and observation period (Fader et al., 2009). An important output of the model is customers' probability of being alive, which represents the probability that a customer remains active. This measure is particularly valuable for CRM because it helps firms distinguish between customers who are likely to continue purchasing and those who may require retention or reactivation efforts.

While the BG/NBD model captures purchase frequency and customer activity status, it does not directly estimate monetary value. Therefore, the Gamma-Gamma model is used to predict customers' expected transaction value. This model captures heterogeneity in spending behavior and provides an estimate of the expected average monetary value for each customer. By combining BG/NBD-based expected purchase frequency with Gamma-Gamma-based monetary value, firms can estimate CLV at the individual customer level (Gou et al., 2013).

Overall, probabilistic CLV models provide a rigorous foundation for customer-level value prediction. By combining BG/NBD and Gamma-Gamma models, firms can estimate both expected purchase frequency and expected monetary value. The inclusion of alive probability further strengthens CLV analysis by capturing customers' relationship status, providing a useful basis for subsequent segmentation based on both future value and likelihood of continued engagement.

However, many studies focus primarily on improving predictive accuracy, while giving less attention to how CLV estimates can support customer classification and CRM decision-making. From a managerial perspective, the ultimate purpose of CLV modeling is not only to predict future value but also to guide marketing actions.

To address this gap, this study uses K-means clustering to classify customers according to predicted six-month CLV and probability of being alive. This approach enables the identification of customer groups with distinct strategic meanings, including lost customers, at-risk customers, potential customers, and VIP customers. Each group requires a different CRM strategy, ranging from selective reactivation and scalable retention to value growth and high-value customer protection.

Therefore, the key contribution of this study lies in connecting probabilistic CLV modeling with actionable customer classification. By integrating BG/NBD, Gamma-Gamma, CLV estimation, and K-means segmentation, the study provides a decision-oriented framework that supports targeted marketing, customer retention, and resource allocation. This extends CLV analysis beyond prediction and demonstrates how customer-level analytics can be transformed into practical CRM strategies.

3. Methodology

3.1. Data

This study uses the Retail Transactions Dataset obtained from Kaggle, which simulates retail purchasing behavior for analytical tasks such as customer segmentation, demand forecasting, and customer value prediction. The raw dataset contains 1,000,000 transaction records and 13 variables, including transaction ID, transaction date, customer name, product, total items purchased, total transaction value, payment method, city, store type, discount application, customer category, season, and promotion status.

Initial data inspection shows that the key modeling fields are complete: `Customer_Name`, `Date`, `Total_Cost`, and `Total_Items` have no missing values. There are also no transactions with non-positive `Total_Cost` or non-positive `Total_Items`. The transaction dates range from 1 January 2020 to 18 May 2024, and the dataset contains 329,738 unique customer-name values before the calibration and repeat-customer filters are applied.

For probabilistic CLV modeling, the transaction-level data are transformed into customer-level behavioral summaries using recency, frequency, observation period, and average monetary value. The final six months of the dataset are reserved as a holdout period, from 18 November 2023 to 18 May 2024,

covering 183 days. The earlier period is used for model calibration. Since the model is estimated with a daily time unit, actual holdout purchase frequency is evaluated using unique customer-day purchase periods rather than raw transaction-row counts.

A key data limitation is that the dataset provides `Customer_Name` rather than a unique customer identifier. Therefore, customer names are used as proxy customer IDs. This may merge different customers with identical names and inflate frequency, monetary value, and CLV for common names. The results should therefore be interpreted as customer-name-level analytical outputs rather than fully verified individual-level customer records.

Another scope limitation is that the BG/NBD and Gamma-Gamma CLV model is fitted only on repeat customers. Before filtering, the calibration dataset contains 308,666 customer-name records. After requiring positive repeat frequency, 139,004 repeat customers are retained, while 169,662 one-time buyers are excluded from the main probabilistic CLV model. This means that the results apply to repeat customers, and one-time buyers would require a separate activation or second-purchase model in future work.

The dataset is suitable for non-contractual CLV modeling because customers can make purchases at any time and do not formally terminate their relationship with the firm. Therefore, customer inactivity is not directly observed and must be inferred from transaction history. This setting justifies the use of the BG/NBD model to estimate expected future purchase behavior and probability of being alive. However, because the dataset does not include gross margin, campaign cost, cost-to-serve, or campaign response data, the estimated CLV in this study should be interpreted as revenue-based or transaction-value-based CLV rather than exact profit-based CLV.

3.2. Research process

Figure 1 presents the research process, which consists of four main stages. First, transaction-level data are cleaned, split into calibration and holdout periods, and transformed into customer-level behavioral indicators, including frequency, recency, observation period, and monetary value. Second, the BG/NBD model is applied to estimate expected future purchase periods and probability of being alive. Third, the Gamma-Gamma model is used to estimate expected average transaction value, which is combined with BG/NBD outputs to calculate six-month revenue-based CLV. Finally, K-means clustering is used to classify customers into distinct CRM segments based on predicted CLV and p_{alive} , followed by holdout validation and strategy development.

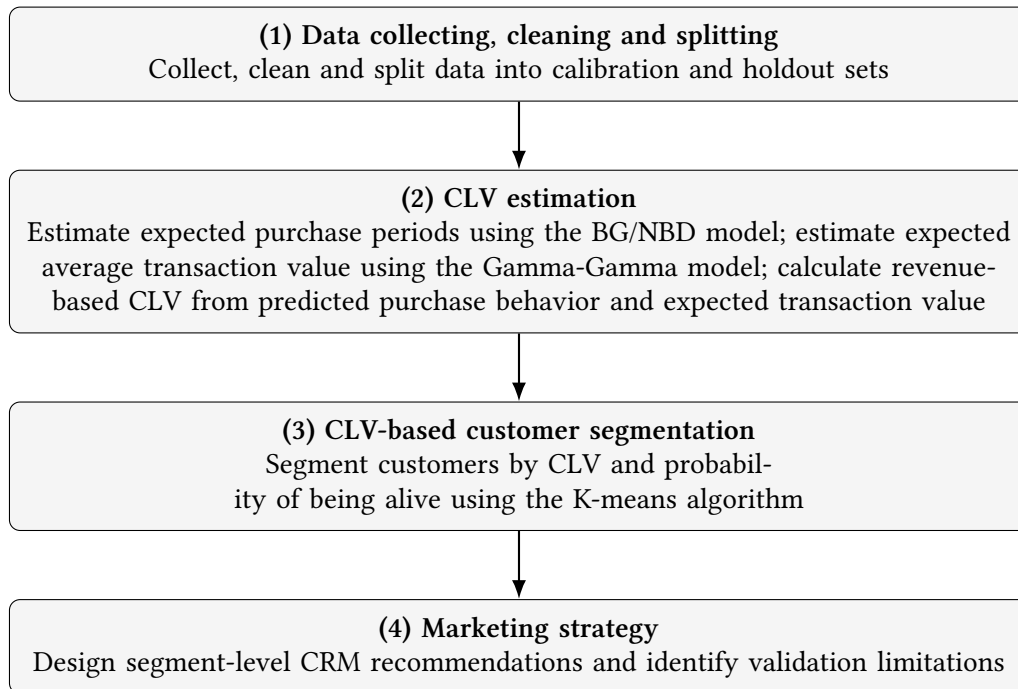


Figure 1: Research procedure

Source: Authors' design

3.3. BG/NBD model

The BG/NBD model is employed to estimate repeat purchase behavior in a non-contractual setting. In this context, customers may purchase repeatedly over time, but the firm cannot directly observe whether a customer has permanently churned. The BG/NBD model addresses this issue by jointly modeling two processes: the transaction process and the dropout process. The model is based on four main assumptions:

H1: When the customer is active, the transaction volume of each customer obeys the Poisson distribution of the transaction rate λ ; that is, the interval of transaction time obeys the Poisson distribution of the transaction rate λ . The formula is as Eq. (1):

$$f(t_j|t_{j-1}; \lambda) = \lambda e^{-\lambda(t_j - t_{j-1})}, t_j > t_{j-1} \geq 0 \quad (1)$$

H2: The probability density function of nonuniformity of customer transaction rate λ obeys *gamma* distribution. The formula is as Eq. (2):

$$f(\lambda|r, \alpha) = \frac{\alpha^r \lambda^{r-1} e^{-\lambda\alpha}}{\Gamma(r)}, \lambda > 0 \quad (2)$$

H3: After each transaction, the probability that the customer becomes silent is p , and the customer churn point follows the binomial distribution. The formula is as Eq. (3):

$$P = p(1 - p)^{j-1}, j = 1, 2, 3, \dots \quad (3)$$

H4: The nonuniformity probability density function of probability p follows a *beta* distribution. The formula is as Eq. (4):

$$f(p|a, b) = \frac{p^{a-1}(1-p)^{b-1}}{B(a, b)}, 0 \leq p \leq 1 \quad (4)$$

where $B(a, b)$ is a beta function, which can be expressed by the gamma function. The formula is as Eq. (5):

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \quad (5)$$

Based on the above assumptions, there are four key expressions for predicting customer purchase behavior:

- (1) The probability $P(X(t))$ of one purchase in a period of time with length x , and the formula is as Eq. (6):

$$P(X(t) = x | r, \alpha, a, b) = \frac{B(a, b+x)}{B(a, b)} \frac{\Gamma(r+x)}{\Gamma(r)x!} \left(\frac{\alpha}{\alpha+t}\right)^r \left(\frac{t}{\alpha+t}\right)^x \\ + \delta_{x>0} \frac{B(a+1, b+x-1)}{B(a, b)} \times \left[1 - \left(\frac{\alpha}{\alpha+t}\right)^r \left\{ \sum_{j=0}^{x-1} \frac{\Gamma(r+j)}{\Gamma(r)j!} \left(\frac{t}{\alpha+t}\right)^j \right\}\right]. \quad (6)$$

- (2) The expected number of transactions $\mathbb{E}[X(t)]$ in a period of t , and the formula is as Eq. (7):

$$\mathbb{E}(X(t) | r, \alpha, a, b) = \frac{a+b-1}{a-1} \left[1 - \left(\frac{\alpha}{\alpha+t}\right)^r {}_2F_1\left(r, b; a+b-1; \frac{t}{\alpha+t}\right)\right]. \quad (7)$$

- (3) The expected number of transactions $\mathbb{E}[Y(t)]$ of each customer in a period of t , and the formula is as Eq. (8):

$$\begin{aligned} \mathbb{E}(Y(t) \mid s, \beta, a, b, x, t_x, T) &= \frac{a + b + x - 1}{1 + \sigma(\beta + T/\beta + t_x)^{s+x}} \left[1 - \left(\frac{\beta + T}{\beta + T + t} \right)^{s+x} \right. \\ &\quad \left. \times {}_2F_1^2 \left(s + x, b + x; a + b + x + 1; \frac{t}{\beta + T + t} \right) \right]. \end{aligned} \quad (8)$$

- (4) The maximum likelihood function of the four parameters for calculating the number of transactions, and the formula is as Eq. (9):

$$\begin{aligned} \mathcal{L}(s, \beta, a, b \mid x, t_x, T) &= \frac{B(a, b + x)\Gamma(s + x)d}{B(a, b)\Gamma(s)(\beta + T)^{s+x}} + \delta_{x>0} \frac{B(a + 1, b + x - 1)\Gamma(s + x)d}{B(a, b)\Gamma(s)(\beta + t_x)^{s+x}} \\ &= A_1 \cdot A_2 \cdot (A_3 + \delta_{x>0}A_4). \end{aligned} \quad (9)$$

where, and the specific calculation formulas are shown in Eqs. (10)–(13):

$$A_1 = \frac{\Gamma(r + x)\alpha^r}{\Gamma(r)} \quad (10)$$

$$A_2 = \frac{\Gamma(a + b)\Gamma(b + x)}{\Gamma(b)\Gamma(a + b + x)} \quad (11)$$

$$A_3 = \left(\frac{1}{\alpha + T} \right)^{r+x} \quad (12)$$

$$A_4 = \left(\frac{a}{b + x - 1} \right) \left(\frac{1}{\alpha + t_x} \right)^{r+x} \quad (13)$$

The parameters are estimated by maximizing the log-likelihood function. In this study, the BG/NBD model is used to estimate expected repeat purchase periods over the six-month holdout and prediction horizon, together with each customer's probability of being alive. This output is particularly important for customer relationship management because it distinguishes customers who are likely to remain active from those who may require retention or reactivation efforts.

3.4. Gamma-Gamma model

While the BG/NBD model estimates how frequently customers are expected to purchase, it does not capture how much customers are expected to spend. Therefore, the Gamma-Gamma model is applied to estimate customers' expected average transaction value. In this study, monetary value refers to transaction value measured by `Total_Cost`; it does not represent profit because product margin and cost-to-serve are unavailable.

The Gamma-Gamma model assumes that each customer has an unobserved average spending level and that observed transaction values vary around this individual-specific mean. It is estimated using customers with repeat purchase histories and positive monetary values. A key assumption of the model is that purchase frequency and monetary value are independent. This assumption is evaluated empirically before using the monetary estimate in the CLV calculation. The model's holdout performance is also treated as a diagnostic check, because the purpose of this study is segmentation and prioritization rather than exact individual-level monetary forecasting.

The average purchase amount of individual customers follows the *gamma* distribution. The formula is Eq. (14):

$$f(m_x \mid p, v, x) = \frac{(vx)^{px} m_x^{px-1} e^{-vxm_x}}{\Gamma(px)} \quad (14)$$

where x is the number of purchases in customer history, p is the scale parameter of the Gamma distribution, and v is the shape parameter of the Gamma model.

Due to the individual to heterogeneity between customers, the shape parameter v follows the *Gamma* distribution with scale parameter q and shape parameter γ . The formula is as Eq. (15):

$$f(m_x | p, q, \gamma, x) = \frac{\Gamma(px - q)}{\Gamma(px)\Gamma(q)} \frac{\gamma^q m_x^{px-1} x^{px}}{(\gamma + m_x x)^{px+q}} \quad (15)$$

The distribution function of the expected transaction amount m_x is obtained from the Bayesian posterior probability. The formula is as Eq. (16):

$$g(v | p, q, \gamma, m_x, x) = \frac{(\gamma + m_x x)^{px+q} v^{px+q-1} e^{-v(\gamma+m_x x)}}{\Gamma(px + q)} \quad (16)$$

In summary, the expected transaction amount of customers in transaction behavior. The formula is as Eq. (17):

$$\mathbb{E}(v | p, q, \gamma, m_x, x) = \frac{(\gamma + m_x x)p}{px + q - 1} = \left(\frac{q - 1}{px + q - 1} \right) \frac{\gamma p}{q - 1} + \left(\frac{px}{px + q - 1} \right) m_x. \quad (17)$$

Finally, the expected purchase frequency from BG/NBD and the expected average transaction value from Gamma-Gamma are combined to estimate revenue-based CLV.

3.5. Revenue-based customer lifetime value

The predicted value of CLV is obtained as the discounted sum of expected future transaction values, where expected purchase behavior is estimated using the BG/NBD model and expected average transaction value is derived from the Gamma-Gamma model:

$$CLV = \sum_{t=1}^n \frac{\mathbb{E}[X(t)] \cdot \mathbb{E}[M]}{(1 + r)^t} \quad (18)$$

where r is the discount rate applied to each prediction period and t is the time period within the prediction horizon.

In this study, CLV is predicted over the next six months using a monthly discount rate of 1%. The discount rate is applied to convert expected future transaction values into present-value terms. Since the available monetary input is `Total_Cost` and the dataset does not include product margin, campaign cost, or cost-to-serve, the resulting CLV is interpreted as revenue-based or transaction-value-based CLV. It is useful for ranking and segmenting customers, but it should not be interpreted as exact profit-based CLV.

3.6. K-Means clustering

To transform CLV predictions into actionable insights, this study applies K-means clustering using two standardized variables: predicted six-month revenue-based CLV and probability of being alive. Standardization prevents the larger numerical scale of CLV from dominating the clustering process. The main analysis uses $K = 4$ to create four CRM-actionable segments: Lost, At-risk, Potential, and VIP. This choice is supported by business interpretability rather than by a purely mechanical selection of the highest silhouette score. Elbow and silhouette diagnostics are used as robustness references, while the final segmentation is assessed by whether the clusters produce meaningful differences in actual holdout activity and transaction value.

4. Results

4.1. Model estimation and prediction result

Table 1 reports the estimated BG/NBD parameters, along with standard errors and 95% confidence intervals. The estimates of r and α indicate heterogeneity in customers' purchase rates, with the relatively large α suggesting that repeat purchases occur infrequently on average. For the dropout process, the parameters a and b suggest variation in customers' inactivity risk, although the overall

dropout tendency remains relatively limited. The narrow confidence intervals further indicate stable parameter estimation. These parameter estimates provide the basis for subsequent p_{alive} and CLV prediction, but model usefulness is evaluated further using holdout validation rather than parameter estimates alone.

Table 1: Value and confidence interval distribution of BG/NBD model parameters

	Coef	Se (coef)	Lower 95% bound	Upper 95% bound
r	1.668658	0.008328	1.652335	1.684980
α	395.606303	2.213100	391.268627	399.943979
a	0.051622	0.001538	0.048608	0.054635
b	0.255405	0.008034	0.239658	0.271152

Source: Authors' calculation

Figure 2 presents the BG/NBD-based retention probability map. The results show that customer survival probability is mainly driven by recency, with recently active customers exhibiting a higher likelihood of remaining alive. In contrast, customers with low recency display consistently low p_{alive} , even when their historical purchase frequency is high. This finding suggests that recent engagement is a stronger indicator of customer retention than purchase frequency alone, supporting the use of p_{alive} for subsequent customer segmentation.

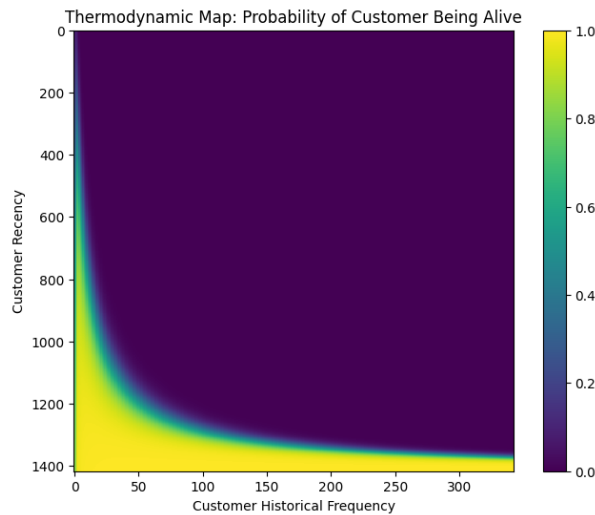


Figure 2: Thermodynamic diagram of customer retention probability

Source: Authors' calculations

Figure 3 shows the distribution of customers' probability of being alive estimated by the BG/NBD model. The distribution is highly concentrated near 1, indicating that a large proportion of customers are predicted to remain active. However, the presence of customers with lower survival probabilities suggests a distinct at-risk group. These results highlight the importance of incorporating p_{alive} into segmentation to distinguish active, potentially loyal customers from those requiring retention efforts.

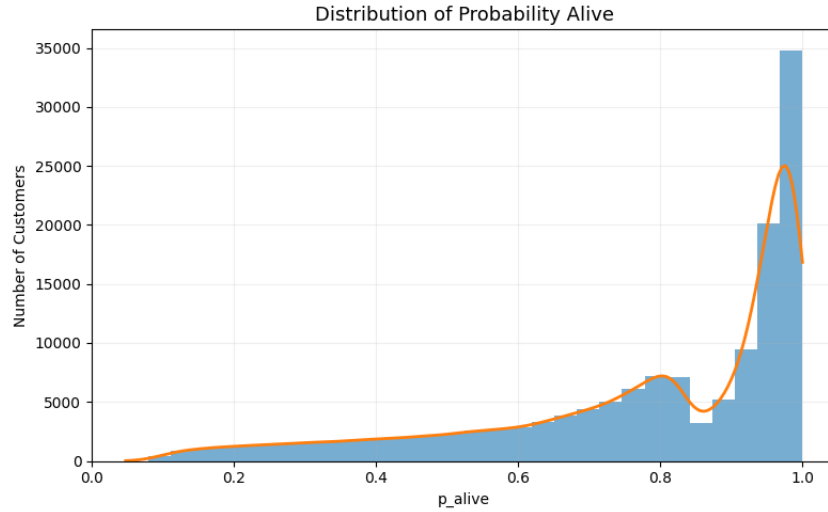


Figure 3: Customer survival curve

Source: Authors' calculation

Figure 4 compares the BG/NBD model predictions with actual holdout repeat purchase periods. The model captures the overall declining pattern of repeat purchase behavior, where most customers make zero or one repeat purchase period and only a small proportion purchase more frequently. Although some deviations are observed, particularly at low transaction counts, the predicted distribution broadly aligns with the empirical pattern. This indicates that the BG/NBD model provides an acceptable directional fit for forecasting repeat purchasing behavior.

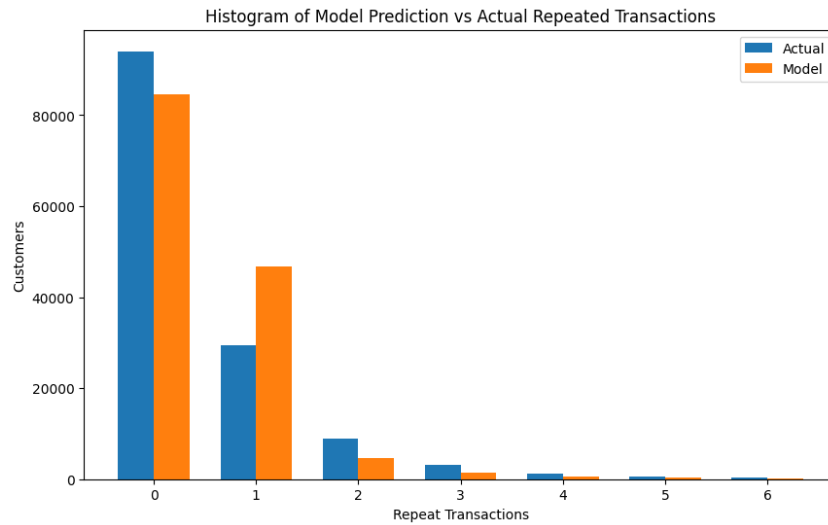


Figure 4: Histogram of BG/NBD prediction and actual holdout purchase periods

Source: Authors' calculation

To evaluate predictive performance more directly, the BG/NBD predictions are compared with actual repeat purchase periods in the six-month holdout period. Because the model is estimated with `freq = 'D'`, the holdout target is measured as unique customer-day purchase periods rather than raw transaction rows.

Table 2: BG/NBD holdout validation for six-month repeat purchase periods

Metric	Value
MAE	0.5659

Table 2: BG/NBD holdout validation for six-month repeat purchase periods

Metric	Value
RMSE	0.8133
Bias mean(predicted - actual)	0.0458
Correlation with actual	0.7713

Source: Authors' calculation

Table 2 shows that the BG/NBD model has a low average prediction error and a strong positive correlation with actual holdout purchase periods. The mean prediction bias is small and positive, indicating mild overprediction on average. To check whether the probabilistic model adds value beyond a simple rule, Table 3 compares BG/NBD with a naive historical-rate baseline.

Table 3: BG/NBD comparison with naive historical-rate baseline

Model	MAE	RMSE	Bias	Correlation
BG/NBD	0.5659	0.8133	0.0458	0.7713
Naive historical-rate baseline	0.5769	0.8329	0.1051	0.7565

Source: Authors' calculation

The BG/NBD model performs slightly better than the naive baseline across MAE, RMSE, bias, and correlation. The improvement is moderate rather than dramatic, but it supports the use of BG/NBD as a structured probabilistic model for repeat-customer purchase prediction.

The probability of being alive is also validated by ranking customers into deciles. Table 4 compares the lowest and highest p_{alive} deciles.

Table 4: Holdout validation of probability of being alive

p_{alive} decile	Customers	Avg. predicted p_{alive}	Actual active rate	Avg. actual purchase periods
Lowest decile	13,901	0.2744	0.1714	0.1993
Highest decile	13,901	0.9931	0.7689	2.2406

Source: Authors' calculation

Customers in the highest p_{alive} decile are much more likely to purchase in the holdout period than customers in the lowest decile. This supports the use of p_{alive} as an activity-ranking variable in the segmentation stage.

Table 5 reports the estimated parameters of the Gamma-Gamma model, including standard errors and 95% confidence intervals. The estimates of p , q , and v are precisely estimated, as indicated by their small standard errors and narrow confidence intervals. However, parameter precision alone does not guarantee strong individual-level monetary prediction; therefore, additional assumption and holdout diagnostics are reported below.

Table 5: Value and confidence interval distribution of Gamma-Gamma model parameters

	Coef	Se (coef)	Lower 95% bound	Upper 95% bound
p	4.272184	0.013151	4.246408	4.297959
q	0.769180	0.002530	0.764220	0.774140
v	4.070424	0.013269	4.044417	4.096432

Source: Authors' calculation

The Gamma-Gamma model requires purchase frequency and monetary value to be approximately independent. In this dataset, the Pearson correlation between frequency and monetary value is 0.0077, and the Spearman correlation is 0.0038, indicating that this assumption is acceptable. However, the holdout diagnostic for expected average transaction value shows limited individual-level accuracy, with MAE = 24.3369, RMSE = 30.5871, WAPE = 46.5427%, and correlation = -0.0096. This means that

the expected average transaction value should not be interpreted as an exact customer-level forecast. Instead, it is used as a smoothed monetary component in a revenue-based CLV ranking and segmentation framework.

Table 6 presents the predicted future purchase behavior and six-month CLV for the highest-value customers. The top customer-name records exhibit very high p_alive values, mostly above 0.99, indicating a strong likelihood of continued activity. Their monetary expectations are relatively stable, ranging around 55–61, suggesting that differences in revenue-based CLV are mainly driven by predicted purchase frequency rather than transaction value.

Table 6: Top 10 customer-name records by predicted six-month revenue-based CLV

Customer Name	Predicted purchase	p_alive	Monetary expectation	CLV (6m)
Michael Smith	34.04	0.9998	60.00	1972.74
James Smith	27.73	0.9997	60.55	1621.86
Michael Johnson	26.55	0.9997	61.03	1565.18
Christopher Smith	24.93	0.9995	59.58	1434.88
David Smith	24.14	0.9997	58.02	1353.09
Michael Jones	23.05	0.9996	59.10	1315.91
Michael Williams	24.48	0.9995	55.23	1305.78
John Smith	23.97	0.9989	55.08	1275.35
Jennifer Smith	22.53	0.9990	57.99	1261.91
Robert Smith	21.89	0.9998	56.72	1199.32

Source: Authors' calculation

The top-ranked records should be interpreted with caution because the dataset uses customer names rather than verified unique customer IDs. Common names may combine multiple real individuals, which can inflate predicted purchase frequency and CLV. This limitation is explicitly considered when interpreting customer-level rankings.

4.2. Customer segmentation

Table 4 presents the K-means clustering results based on predicted six-month revenue-based CLV and p_alive . The four-cluster solution produces distinct customer groups with different levels of expected transaction value and relationship status, namely Lost, At-risk, Potential, and VIP customers.

Table 7: Clustering results by K-means method and group profiling

Cluster	Group	Average p_alive	Average CLV (6m)	Sample size
1	Lost	0.3525	7.2788	22,460 (16.16%)
2	At risk	0.7250	16.6527	44,568 (32.06%)
3	Potential	0.9544	41.6638	70,073 (50.41%)
4	VIP	0.9961	285.5852	1,903 (1.37%)

Source: Authors' calculation

The VIP segment contains 1,903 customers, accounting for 1.37% of the repeat-customer modeling sample. Despite its small size, this group records the highest average p_alive (0.9961) and the highest average six-month revenue-based CLV (285.59). This indicates that VIP customers are highly likely to remain active and are expected to generate substantially higher future transaction value than other segments.

The Potential segment includes 70,073 customers, representing 50.41% of the sample. This group shows a very high average p_alive (0.9544) and a moderate average six-month CLV (41.66). Potential customers are therefore likely to remain active but still have room for value growth through upselling, cross-selling, and loyalty upgrade campaigns.

The At-risk segment consists of 44,568 customers, or 32.06% of the sample. Its average p_alive

is 0.7250, but its average six-month CLV is only 16.65. This indicates that many customers in this segment may still purchase, but their expected future transaction value is limited. The appropriate strategy is therefore scalable and cost-controlled retention rather than expensive one-to-one intervention for all customers.

The Lost segment contains 22,460 customers, accounting for 16.16% of the sample. This group has the lowest average p_alive (0.3525) and the lowest average six-month CLV (7.28), indicating weak future engagement and limited expected value. Intensive marketing investment in this group may be inefficient, and only low-cost selective reactivation should be considered.

Figure 5 illustrates the K-means customer segmentation based on six-month revenue-based CLV and p_alive . The results show that customer value is highly unequal across the repeat-customer base. More importantly, high activity probability does not always imply high expected transaction value. Therefore, combining CLV with p_alive provides a more meaningful basis for customer classification than using either variable alone.

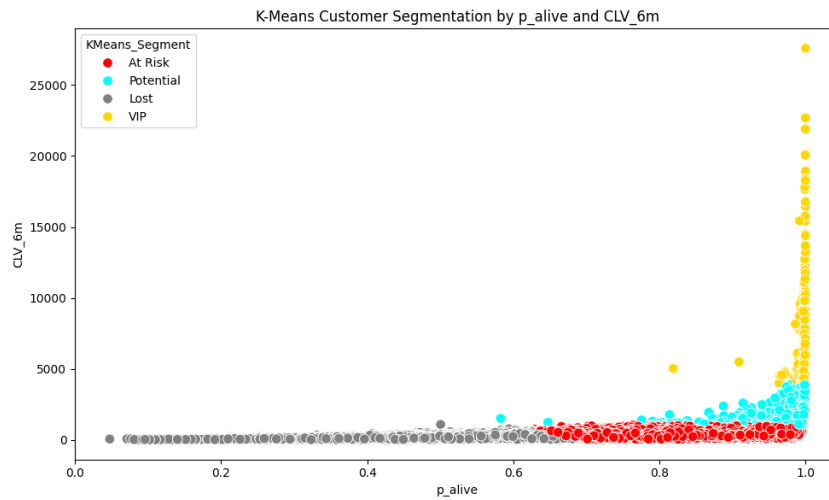


Figure 5: Customer segmentation by K-Means

Source: Authors' calculation

To evaluate whether the segments are meaningful beyond the model outputs, Table 8 compares each segment's actual behavior in the six-month holdout period.

Table 8: Segment-level holdout validation

Segment	Customers	Avg. predicted CLV	Actual active rate	Avg. actual value	Total actual value
VIP	1,903	285.59	0.9937	356.66	678,722.16
Potential	70,073	41.66	0.4356	37.27	2,611,843.24
At Risk	44,568	16.65	0.1937	12.02	535,883.37
Lost	22,460	7.28	0.1736	10.68	239,840.54

Source: Authors' calculation

The segment-level validation supports the business usefulness of the segmentation. VIP customers have the highest actual active rate and the highest average actual holdout value. Potential customers rank second in both predicted CLV and actual average value, while At-risk and Lost customers show substantially weaker holdout behavior. This result is important because the Gamma-Gamma model has limited individual-level monetary prediction accuracy. The stronger conclusion is therefore not that the model perfectly forecasts each customer's future spending, but that it creates meaningful customer groups for CRM prioritization.

A K-selection diagnostic was also conducted using elbow, silhouette, and cluster-profile comparisons. The diagnostic shows that alternative values such as $K = 3$ may be competitive in geometric separation, but $K = 4$ is retained because it provides a clearer CRM action framework: Lost, At-risk,

Potential, and VIP. With $K = 3$, the segmentation becomes too coarse for differentiated strategy; with $K = 5$, the VIP group splits into very small high-value subgroups that are less stable for a general CRM framework. Therefore, $K = 4$ is selected primarily for interpretability and actionability, while technical clustering scores are used as diagnostic references rather than the sole decision rule.

4.3. Marketing strategy

The segmentation results provide a practical basis for differentiated CRM strategy. Following the principle of CLV-based resource allocation, marketing resources should not be distributed equally across all customers. Instead, firms should allocate budgets, service efforts, and promotional incentives according to customers' expected future transaction value, likelihood of continued engagement, and potential responsiveness to marketing actions (Kumar, 2010). This logic is consistent with value-based customer management, in which firms classify customers based on expected lifetime value and design segment-specific strategies rather than applying a standardized campaign to the entire customer base (Kim et al. 2006).

Based on the four customer segments, this study proposes a protect-grow-stabilize-recover CRM strategy. The proposed allocation ranges are illustrative rather than optimized budget prescriptions, but as a strategic starting point that should be validated through A/B testing, uplift modeling, and incremental CLV analysis before implementation.

4.3.1. VIP group

The firm may allocate approximately 20-25% of its CRM or retention marketing budget to the VIP segment, despite its small size. This allocation reflects the segment's disproportionately high CLV and its strategic importance to future revenue protection.

For VIP customers, the marketing objective should not focus primarily on price-based incentives. Since these customers already show high engagement and strong purchase potential, excessive discounting may reduce net return without generating substantial loyalty gains. Instead, resources should be directed toward relationship-enhancing initiatives, such as personalized cross-selling and upselling, premium customer experiences, priority service, exclusive benefits, and advocacy programs.

Based on this logic, the proposed VIP campaigns are designed to protect existing high-value customers while expanding their future contribution through deeper personalization, stronger relationship management, and value-added experiences.

Campaign 1: "VIP Care & Privilege Program"

The first proposed campaign focuses on proactive retention and differentiated service treatment for high-value customers. This campaign is grounded in the literature on customer relationship management and relational benefits. Gwinner et al. (1998) shows that long-term customer relationships generate several relational benefits, including confidence benefits, social benefits, and special treatment benefits. These findings support the idea that VIP customers should receive not only economic incentives, but also recognition, priority service, and exclusive treatment. In practice, the campaign should include a dedicated care program for VIP customers, such as priority customer support, faster response time, birthday gifts, personalized privileges, early access to new products, and access to premium services. The focus should not be deep price discounts, but rather the creation of a sense of recognition, exclusivity, and superior customer experience. The firm can implement this campaign by tagging VIP customers in the CRM system so that their transactions, complaints, and service requests are automatically prioritized. For customers with exceptionally high CLV, the company may assign a dedicated customer service representative or account owner to monitor their behavior and provide individualized support. The objective of this campaign is to increase the retention rate, improve customer satisfaction, and reduce the risk of VIP customers switching to competitors. Since VIP customers have high p-alive and high expected future value, proactive relationship management is more appropriate than reactive promotion-based marketing.

Campaign 2: "Next Best Offer" - Increasing Purchase Value:

The second proposed campaign aims to increase the purchase value of VIP customers through personalized upselling and cross-selling. This campaign is supported by research on next-product-to-buy models and customer-level targeting. Knott et al. (2002) argue that next-product-to-buy models can improve cross-selling effectiveness by predicting which product a customer is most likely to pur-

chase next, thereby reducing waste from poorly targeted marketing activities. For the VIP segment, the firm should use historical purchase data to recommend products, service packages, or offers that are most relevant to each customer's previous behavior. Since this group already has high CLV and strong purchase potential, the campaign should prioritize upselling and cross-selling rather than mass discounting. Examples include recommending premium versions of products, complementary bundles, subscription plans, service upgrades, or personalized offers based on past purchasing patterns. The communication message should be highly personalized, such as: *"This premium package matches your previous usage pattern"* or *"An exclusive offer selected for our VIP customers."* The objective of the campaign is to increase average order value, stimulate repeat purchases, and further improve the customer's CLV over the next six months.

Campaign 3: "VIP Recovery Trigger" - Proactive Retention Before Churn

The third campaign focuses on early intervention when VIP customers show signs of declining engagement. This campaign is based on the literature on churn prediction and defection detection. Neslin et al. (2006) emphasize the importance of detecting customer defection risk through predictive churn models, which help firms identify customers who are likely to leave before actual churn occurs. The campaign should involve setting up early-warning triggers when a VIP customer shows abnormal behavior, such as a long period without purchase, a decline in order value, lower response to marketing messages, reduced engagement, or unresolved complaints. Once these signals are detected, the CRM or customer service team should proactively contact the customer to understand the reason for the decline, resolve potential issues, and provide an appropriate recovery offer. For VIP customers, the recovery approach should be more personal than standard automated messages. Instead of sending generic mass emails or SMS messages, the company should prioritize direct calls, personalized emails, or one-to-one service recovery actions. The main objective is to reduce churn risk, address negative experiences early, and protect the future revenue generated by high-value customers.

Campaign 4: "VIP Referral & Ambassador Program" - VIP Customers into Advocates

The final proposed campaign aims to transform satisfied VIP customers into customer advocates and referral sources. This campaign is supported by previous research (Schmitt, 2011) on referral programs and customer value. Their study finds that referred customers tend to have higher retention and higher average value than comparable non-referred customers, with the average value of referred customers being at least 16% higher. In this campaign, the firm can create personalized referral codes for each VIP customer and offer benefits to both the referrer and the referred customer. The campaign can be designed as a "VIP refers VIP" program, encouraging high-value customers to introduce similar high-quality customers from their social or professional networks. In addition, selected VIP customers can be invited to participate in reviews, testimonials, case studies, or brand community activities to strengthen brand credibility. The objective of this campaign is twofold. First, it helps the company acquire new customers with potentially high value and stronger retention potential. Second, it deepens the emotional and relational attachment of existing VIP customers by recognizing them not only as buyers, but also as trusted brand advocates. Therefore, the referral campaign contributes to both customer acquisition quality and VIP relationship reinforcement.

In summary, for the VIP segment, firms should adopt a "protect and expand" strategy. The protect objective focuses on retaining high-value customers through priority service, proactive care, and differentiated experiences. The expand objective aims to increase future value through personalization, cross-selling, upselling, and referral programs. Therefore, VIP customers should be managed as strategic assets rather than targets for mass discounts. This approach helps the firm protect existing high-value revenue while further increasing the CLV of its highest-value customers.

4.3.2. Potential group

This research proposes a "grow and upgrade" strategy for the Potential segment. While their individual CLV is lower than VIPs, their large base and high probability of remaining active indicate significant expansion potential. The strategic goal is to increase purchase frequency, basket size, and premium adoption, justifying a substantial 30–35% allocation of the CRM budget. Even a moderate improvement in this segment can drive massive aggregate revenue growth and create a pipeline for future VIPs.

Campaign 1: “CRM Lifecycle Upsell & Cross-sell”

The first proposed campaign should account for approximately 40% of the resources allocated to the Potential segment. This campaign is based on customer lifecycle management and next-best-offer logic. A McKinsey case on targeted online marketing shows that cross-selling and category-penetration techniques helped an online retailer increase yearly sales by about 20% and annual EBITDA by roughly 30% after broader implementation. The same case also reported increases in email conversions, on-site conversions, and marketing ROI. In this campaign, the CRM system should automatically identify relevant upsell or cross-sell opportunities after each transaction. Within three to five days after purchase, potential customers should receive personalized product recommendations, upgraded service packages, complementary bundles, or subscription offers. This timing is appropriate because customers are still close to the purchase context and may be more receptive to related offers. The key message should not be framed as a generic promotion. Instead, it should emphasize relevance and personalization, such as: “Based on your recent purchase, this upgraded package may better match your needs.” This approach is also supported by McKinsey’s personalization research, which reports that 71% of consumers expect personalized interactions, while faster-growing companies generate 40% more of their revenue from personalization than slower-growing firms. The main objectives of this campaign are to increase average order value, improve cross-sell conversion rate, and raise the six-month CLV of potential customers.

Campaign 2: “VIP Progress Accelerator”

The second proposed campaign should receive approximately 30% of the Potential segment budget. This campaign uses a loyalty progress mechanism to encourage customers to increase spending and move closer to VIP status. The theoretical foundation of this campaign comes from the goal-gradient hypothesis. Kivetz et al. (2006) show that customers tend to increase effort as they approach a reward, and their study applies this principle to reward programs and customer retention. In practice, the firm can design a visible progress bar in the customer account, mobile app, or CRM communication. For example, the system may display: “Only USD X more to unlock VIP benefits.” This message makes the customer’s progress toward a higher-value tier more visible and psychologically motivating. Rather than offering immediate discounts, the firm encourages customers to increase spending to obtain a higher membership status and exclusive benefits. The objective of this campaign is to increase basket size, stimulate repeat purchases, and convert high-potential customers into VIP customers. This campaign is especially suitable for the Potential segment because these customers are already highly active, but their current monetary value is still below the VIP level.

Campaign 3: “Proactive 1-to-1 Advisory Care”

The third proposed campaign should account for approximately 20% of the segment budget. Unlike standard promotional messages, this campaign focuses on relationship-building and personalized consultation. For Potential customers who have not made a transaction for 45–60 days, the customer service or CRM team should proactively contact them to understand their needs, provide support, and recommend suitable products or service upgrades. The purpose of this campaign is not to create sales pressure, but to establish a personalized advisory relationship. For example, customer service staff may ask whether the previous product or service met the customer’s expectations, suggest complementary solutions, or introduce a premium package based on the customer’s previous purchase history. This type of interaction can make upselling feel more relevant and helpful rather than aggressive. This campaign is also consistent with the broader evidence on personalization. McKinsey emphasizes that customers increasingly expect companies to recognize their individual needs, and poor personalization can lead to frustration and customer switching. Therefore, proactive one-to-one advisory care can support both customer relationship development and future value expansion. The main objectives of this campaign are to increase customer satisfaction, strengthen trust, and create a stronger foundation for future upselling and cross-selling.

Campaign 4: “Potential Exclusive Access”

The final proposed campaign should receive the remaining 10% of resources. This campaign aims to strengthen emotional engagement and make potential customers feel closer to the VIP tier. It may include invitation-only events, early access to new products, premium product trials, personal-

ized demonstrations, or exclusive online communities. This campaign is supported by BCG's research on personalization. BCG reports that approximately four-fifths of surveyed consumers worldwide are comfortable with personalized experiences, and most expect firms to provide them. The same research identifies value, enjoyment, and convenience as the three most frequently cited benefits of personalization. In addition, BCG's research on loyalty programs suggests that customers increasingly expect loyalty programs to provide differentiated experiences beyond monetary rewards, including personalized benefits, free content, and relevant partnerships. For the Potential segment, exclusive-access campaigns can create a stronger sense of recognition and aspiration. These customers are not yet VIPs, but they are active and valuable enough to be encouraged toward a higher-value relationship with the firm. Therefore, this campaign should be used to increase perceived exclusivity, deepen engagement, and support future VIP conversion. In summary, the Potential segment should receive growth-oriented investment. By encouraging these customers to increase spending and move toward the VIP tier, the firm can build a stronger pipeline of future high-value customers and improve overall customer lifetime value.

4.3.3. At-risk group

For the At-risk segment, the firm may allocate approximately 25-30% of its CRM and retention marketing budget. Although the At-risk segment accounts for a substantial share of the customer base, it is smaller than the Potential segment and has a much lower average CLV. Therefore, the strategic objective is not to invest heavily in all At-risk customers, but to stabilize the most recoverable customers in a cost-efficient manner.

This segment should be managed through a stabilize and recover strategy. Customers with relatively higher CLV, stronger p_alive , or recent signs of reduced engagement should receive more personalized recovery actions. In contrast, customers with low expected value or weak response probability should be managed through automated, low-cost campaigns. This approach allows the firm to prevent further value decline while avoiding excessive spending on customers with limited future transaction value. For implementation, this study proposes the marketing campaign for this group.

Marketing Campaign: "Come Back & Save"

The campaign should use a multi-channel approach, combining CRM automation, personalized incentives, re-targeting advertisements, and mobile reminders. The campaign can be triggered when customers have no transaction for 30–60 days. In the first stage, the CRM system sends a personalized email based on the customer's previous purchase history. For example, customers may receive product reminders, replenishment suggestions, or a moderate return incentive. If the customer does not respond, the second stage activates digital re-targeting ads to maintain brand visibility and remind the customer of relevant products. In the third stage, the firm can send a follow-up message with a limited time offer to create urgency. Finally, mobile channels such as push notifications or Zalo messages can be used as a low-cost final reminder. The level of incentive should depend on the customer's expected value. At-risk customers with higher CLV may receive more personalized offers or stronger incentives, while lower-value customers should receive standardized automated messages. This prevents the firm from spending too much on customers who are unlikely to generate sufficient future value.

Overall, the At-risk strategy focuses on efficient retention at scale. Since this group is large, even a small improvement in repeat purchase behavior can generate meaningful aggregate revenue. However, because average CLV remains modest, marketing investment should be carefully controlled and guided by expected recovery value.

4.3.4. Lost group

For the Lost segment, the firm should allocate only 5–10% of its CRM or retention marketing budget. This group has the lowest average p_alive and six-month CLV, indicating weak future engagement and limited expected transaction value. Therefore, the strategic objective is not broad reactivation, but selective recovery of customers who still show moderate return potential.

Marketing Campaign: "Final Win-back Offer"

The campaign should be designed as a low-cost, highly automated reactivation program. Unlike the At-risk segment, which still deserves broader retention efforts, the Lost segment should not receive expensive personalized service or heavy promotional investment. The campaign begins with

an automated win-back email sent to customers in the recoverable lost group. The message should refer to the customer's previous purchase history and include a simple, time-limited offer, such as free shipping, a moderate discount, or a personalized product recommendation. The offer should be valid for only 5–7 days to create urgency without increasing long-term discount dependency. If the customer does not respond, the firm may send one short reminder through SMS or Zalo OA. Re-targeting ads should be used only for customers with stronger past spending behavior, because paid media can quickly become inefficient for low-CLV customers. BCG and Capillary Technologies' data-led customer engagement framework also emphasizes the importance of using data and automation to engage customers across the journey, rather than relying on a single generic campaign.

Overall, the Lost strategy focuses on budget discipline and selective reactivation. Unlike At-risk customers, Lost customers should not receive broad retention investment. They should only be targeted when the expected recovery value exceeds the cost of intervention.

The proposed strategy follows a value-based allocation logic: protect VIP customers, grow Potential customers, stabilize At-risk customers, and selectively recover Lost customers. By aligning CRM investment with CLV and p_{alive} , firms can improve retention efficiency, increase customer value, and allocate marketing resources more effectively.

5. Discussion

The findings show that combining BG/NBD and Gamma-Gamma models provides a useful framework for repeat-customer ranking and segmentation in a non-contractual retail setting. The BG/NBD model captures differences in purchase frequency and customer activity probability, while the Gamma-Gamma model contributes an expected average transaction value component. Together, these outputs generate a six-month revenue-based CLV indicator that can support CRM prioritization.

The holdout validation provides a more balanced interpretation of model performance. BG/NBD performs slightly better than a naive historical-rate baseline and shows a strong positive correlation with actual holdout purchase periods. The p_{alive} validation also shows clear ranking power: customers in the highest p_{alive} decile are substantially more likely to purchase in the holdout period than customers in the lowest decile. These results support the use of BG/NBD outputs for customer activity prediction and segmentation.

The Gamma-Gamma results should be interpreted more cautiously. The frequency-monetary independence assumption is acceptable because both Pearson and Spearman correlations are close to zero. However, the holdout diagnostic shows weak individual-level accuracy in predicting average transaction value. Therefore, the estimated CLV should not be interpreted as an exact customer-level revenue forecast. It is more appropriate as a revenue-based ranking and segmentation indicator.

The segment-level validation supports this interpretation. VIP and Potential customers show stronger actual holdout behavior than At-risk and Lost customers, especially in actual active rate and average actual transaction value. This indicates that the model is more useful at the group-prioritization level than at the exact individual-forecasting level. From a managerial perspective, the results support a differentiated CRM approach: protect VIP customers, grow Potential customers, stabilize At-risk customers through cost-controlled retention, and selectively recover Lost customers through low-cost reactivation.

6. Conclusion and recommendations

This study applies BG/NBD and Gamma-Gamma models to estimate six-month revenue-based CLV and probability of being alive, then uses K-means clustering to classify repeat customers into four CRM segments: Lost, At-risk, Potential, and VIP. The results show that combining CLV with alive probability provides a more actionable basis for customer segmentation than relying only on historical transaction behavior or CLV alone.

The main contribution of this study is to connect probabilistic CLV modeling with actionable CRM decision-making while explicitly validating the model outputs. BG/NBD holdout validation and p_{alive} decile analysis support the use of purchase-frequency and activity-status estimates. Although the Gamma-Gamma model has limited individual-level monetary prediction accuracy, segment-level holdout validation confirms that the resulting customer groups differ meaningfully in actual future activity and transaction value.

Based on the findings, firms should adopt differentiated CRM strategies across customer groups.

VIP customers should be protected through priority service and personalized relationship management. Potential customers should be targeted with upselling, cross-selling, and loyalty upgrade campaigns. At-risk customers should be managed through scalable and cost-controlled retention programs, whereas Lost customers should receive only selective and low-cost win-back efforts. These recommendations should be treated as strategic decision-support suggestions rather than guaranteed profit-maximizing actions.

This study has several limitations. First, the dataset is simulated and may not fully reflect real customer behavior. Second, the dataset provides `Customer_Name` rather than a verified unique customer ID, which may merge different individuals with identical names. Third, the analysis estimates revenue-based CLV rather than profit-based CLV because product margins, cost-to-serve, campaign costs, and response data are unavailable. Fourth, the main probabilistic CLV model is fitted only on repeat customers, while one-time buyers are excluded and should be modeled separately in future work. Fifth, the Gamma-Gamma model shows limited individual-level accuracy in predicting average transaction value, so CLV should be interpreted primarily as a ranking and segmentation indicator. Finally, the proposed marketing strategies should be validated through A/B testing, uplift modeling, and incremental CLV analysis before operational deployment.

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